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Event-triggered impulsive control for switched delay systems and its application in image encryption of switched neural networks^{*}Zhifeng Lu^a, Haiying Wang^{a,*}, Yujuan Tian^{b,*}^a School of Science, China University of Geosciences (Beijing), Beijing 100083, PR China^b School of Mathematics and Statistics, Shandong Normal University, Ji'nan 250014, PR China

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ABSTRACT

This paper addresses the event-triggered impulsive control problem for nonlinear switched systems with time delay. The main challenge is that the delayed state may originate from a historical interval governed by a subsystem different from the current one, while switching may also occur before the next impulse is generated. To address this issue, a mode-dependent event-triggered mechanism is developed by incorporating an exponential attenuation term associated with the initial historical information. Based on the Lyapunov-Razumikhin method and the average dwell time technique, sufficient criteria are established to ensure exponential stability of the considered system and rule out Zeno behavior. An explicit lower bound on the inter-event times is also obtained, which characterizes the admissible triggering intervals under the combined influence of switching and delay. Since the impulse input is applied only at triggering instants, no continuous control signal needs to be transmitted between two successive triggers, and the communication burden associated with control transmission from the controller to the actuator can be reduced. The proposed theoretical results are further applied to the synchronization of switched neural networks, where the design conditions for the triggering mechanism and impulsive control gain are formulated as linear matrix inequalities. Numerical simulations and an image encryption application demonstrate the effectiveness and practical applicability of the developed method.

1. Introduction

Switched systems, representing a typical category of hybrid dynamical systems, comprise a finite set of subsystems together with a logical law governing their switching (Liberzon, 2023). They provide a natural modeling framework for complex dynamical and engineering systems with multiple operating modes, abrupt structural changes, or logical decision mechanisms, and have been widely used in chaotic systems (Vadivel et al., 2022), neural networks (Cao et al., 2016), stirred tank reactors (Wang et al., 2024), power converters (Beneux et al., 2019), and aero-engines (Wu et al., 2024). In many practical engineering processes, time delay is also unavoidable due to the finite speed of signal propagation, computational processing, and actuator response. The presence of delay makes the system evolution depend strongly on historical states, which may induce oscillation, performance degradation, instability, or even chaotic behavior (Senan et al., 2017; Xing et al., 2025). When switching and delay coexist, the stability analysis and control synthesis become more complicated, since the historical state involved in the delay may

no longer be associated with the currently active subsystem. To address these issues, various control methods have been developed, including sliding mode control (Jiang et al., 2023), prescribed-time control (Yao et al., 2024a), impulsive control (Ren & Xiong, 2019), resilient control (He et al., 2019), and event-triggered control (ETC) (Zhang & Raissi, 2021).

Among these methods, ETC has garnered significant research attention because it updates the control input only when a predefined state-dependent condition is satisfied, thereby reducing unnecessary data transmission and improving resource utilization (Syed Ali et al., 2019; Yang et al., 2025; Yao et al., 2024c). Consequently, this control paradigm has been extensively investigated for switched systems in the contexts of state-feedback control (Zhao et al., 2022; Zobiri et al., 2020), sampled-data control (Xiao et al., 2019), and observer-based output feedback control (Li et al., 2018). However, most ETC strategies for switched systems still require the control signal to be maintained between two consecutive triggering instants. Moreover, to keep consistency between the system mode and the controller mode, the control

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input typically needs to be updated at least once within each switching interval. These requirements inevitably increase the reliance on transmitted data and raise the risk of network congestion, especially for systems with frequent switching. In contrast, impulsive control provides another efficient control strategy, in which the control action is applied only at specific discrete instants. Compared with continuous feedback control, impulsive control has a simpler implementation structure and can reduce the dependence on continuous signal transmission (Chen et al., 2023; Yao et al., 2024b). In view of this, event-triggered impulsive control (ETIC), which combines the merits of ETC and impulsive control, has been proposed and has received considerable attention (Ai et al., 2022; Li et al., 2020; Zhu et al., 2018). Within the ETIC framework, an impulse signal is generated solely upon the activation of an appropriately constructed event-triggered mechanism (ETM). Although real-time monitoring remains necessary to evaluate the triggering condition, ETIC eliminates the need for continuous control signal transmission between successive triggers, thereby providing a more efficient communication pattern. More importantly, the requirement for a forced trigger within each switching interval can be removed, making ETIC particularly suitable for switched systems, even those with frequent switching.

To date, several important results on ETIC for switched systems have been reported. In Zhang et al. (2022), the finite-time stability of switched systems was explored via ETIC, where a mode-dependent event-triggered mechanism (MDET) was constructed using multiple Lyapunov functions. Subsequently, the prescribed-time stability for switched systems within the ETIC framework was established in Zhang et al. (2024) by incorporating tunable exponential parameters into the MDET. More recently, a dynamic MDET was developed in Gao et al. (2025), attaining the input-to-state stability for switched systems through ETIC. These studies have significantly promoted the development of ETIC for switched systems. Nevertheless, they neglect the impact of time delay, and hence the proposed ETMs and stability criteria are only applicable to delay-free switched systems. When time delay is involved, the delayed state may correspond to a historical interval governed by a subsystem different from the current one. Moreover, switching may occur within a triggering interval before the next impulse is generated, causing interactions among the delayed state, the switching topology, and the event-triggered impulse actions. This coupling complicates the construction of an effective triggering threshold and the exclusion of Zeno behavior. Hence, developing ETIC strategies for switched delay systems is nontrivial. To the best of our knowledge, this problem has not yet been systematically explored, even in the linear case.

Inspired by the aforementioned considerations, this paper investigates exponential stability of general nonlinear switched delay systems under ETIC. By employing the Lyapunov-Razumikhin method and the average dwell time (ADT) technique, the joint influence of switching topology, time delay, and impulse actions is captured within a unified theoretical framework. The principal contributions are outlined as follows.

- In contrast to existing ETIC schemes that are applicable only to delay-free switched systems (Gao et al., 2025; Zhang et al., 2024, 2022), this work proposes a novel MDET tailored for switched delay systems. The developed mechanism incorporates an exponential attenuation term that explicitly involves the initial historical information, thereby establishing an analytical connection between the current state and the delayed historical state. This design enables time-delay effects to be addressed within the Lyapunov-Razumikhin framework, which has not been achieved in previous ETIC approaches for switched systems.
- Without imposing any preassigned restrictions on the impulse sequence, sufficient criteria are derived to rule out Zeno behavior and guarantee exponential stability. These criteria establish explicit relationships among the switching signal, the triggering parameters, and the impulse strengths, thereby providing a principled basis for MDET design. Moreover, a lower bound on the inter-event times is obtained, which quantitatively reveals the joint effects of time delay and switching on the admissible triggering intervals, and this, in

turn, enhances the flexibility of impulse sequence generation under the proposed ETIC strategy.

- The developed theoretical results are further applied to the synchronization of switched neural networks (SNNs). By formulating the design conditions for the MDET and the impulsive control gain as linear matrix inequalities (LMIs), feasible synchronization criteria are derived. Building upon this, an image encryption algorithm is presented to illustrate the practical applicability of the proposed ETIC strategy.

The rest of this paper is arranged as follows. Section 2 provides some preliminaries. Section 3 details the MDET with non-Zeno behavior and the stability analysis. Section 4 addresses synchronization of SNNs. Section 5 gives numerical simulations and an image encryption application. Section 6 states the conclusion.

Notations. Let $\mathbb{Z}_+$, $\mathbb{R}$, and $\mathbb{R}_+$ denote the sets of positive integers, real numbers, and nonnegative real numbers, respectively. $\mathbb{R}^n$ represents the n -dimensional real space equipped with the Euclidean norm $\|\cdot\|$. S^T and S^{-1} refer to the transpose and inverse of matrix S , respectively. $S > 0$ ($S < 0$, $S \geq 0$, $S \leq 0$) indicates that S is positive definite (negative definite, positive semi-definite, negative semi-definite). The identity matrix is denoted by I . The maximum of c_1 and c_2 is written as $c_1 \vee c_2$. For any $\mathcal{G}, \mathcal{H} \subseteq \mathbb{R}^m$ ($1 \leq m \leq n$), $\mathbb{PC}(\mathcal{G}, \mathcal{H}) = \{v : \mathcal{G} \rightarrow \mathcal{H} \mid v \text{ is continuous except for finitely many points, and } v(t^+), v(t^-) \text{ exist with } v(t) = v(t^+)\}$. Given $\gamma > 0$, the linear space $\mathbb{PC}_\gamma = \mathbb{PC}([-\gamma, 0], \mathbb{R}^n)$ is equipped with the norm $\|v\|_\gamma = \sup_{\theta \in [-\gamma, 0]} \{ |v(\theta)| \}$. A continuous function $\varpi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ belongs to class $\mathcal{K}$ if ϖ is strictly increasing and $\varpi(0) = 0$. It belongs to class $\mathcal{K}_\infty$ if ϖ further satisfies $\varpi(s) \rightarrow \infty$ as $s \rightarrow \infty$. Sign $\star$ denotes the symmetric block. $\Lambda = \{1, 2, \dots, n\}$.

2. Preliminaries

Consider the nonlinear switched delay system

$$\begin{cases} \dot{x}(t) = f_{\sigma(t)}(x_t), & t \geq t_0, \\ x_{t_0} = \xi, \end{cases} \quad (1)$$

subject to impulses

$$x(t) = g_k(x(t^-)), \quad t = t_k, \quad k \in \mathbb{Z}_+, \quad (2)$$

where $x(t) \in \mathbb{R}^n$ is the system state, $\sigma(t) : [t_0, \infty) \rightarrow \mathcal{I} = \{1, 2, \dots, N\}$ denotes the switching signal. For $t \geq t_0$, $x_t \in \mathbb{PC}_\gamma$ is defined as $x_t(\theta) = x(t + \theta)$ for $\theta \in [-\gamma, 0]$, where $\gamma > 0$ is the time delay. $\xi \in \mathbb{PC}_\gamma$ is the initial function. $\{t_k, k \in \mathbb{Z}_+\}$ (short for $\{t_k\}$) denotes the impulse sequence generated by the MDET. For each $i \in \mathcal{I}$ and $k \in \mathbb{Z}_+$, $f_i : \mathbb{PC}_\gamma \rightarrow \mathbb{R}^n$ and $g_k : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfy $f_i(0) = g_k(0) = 0$. Assume that f_i and g_k satisfy suitable conditions ensuring a unique solution in relevant time intervals (Ren & Xiong, 2019). Moreover, the solution $x(t)$ of system (1) with (2) is right-continuous.

The principal objective of this study is to develop ETIC strategies that achieve exponential stability for system (1) with (2). A block diagram corresponding to this problem is presented in Fig. 1. Once the specified triggering rule is satisfied, the corresponding triggering instant t_k is generated. The state information is subsequently transmitted to the impulse generator, and the system state is updated to $x(t_k)$. Unlike the ETC schemes in Li et al. (2018), Xiao et al. (2019), Zhao et al. (2022), Zobiri et al. (2020), where the control input remains constant between consecutive triggering instants, the proposed approach involves no continuous control transmission. Consequently, despite the need for continuous state monitoring to evaluate the triggering condition, the communication load on the controller-actuator channel is alleviated.

Definition 1 (Hespanha & Morse, 1999). Let $\mathcal{F}$ be a class of switching signals that satisfies

$$N(t, s) \leq \frac{t-s}{\tau_a} + N_0, \quad \forall t \geq s \geq t_0,$$

where $N(t, s)$ is the number of switches on the interval $(s, t]$, $N_0 > 0$ is the chatter bound, and $\tau_a > 0$ is the ADT constant.

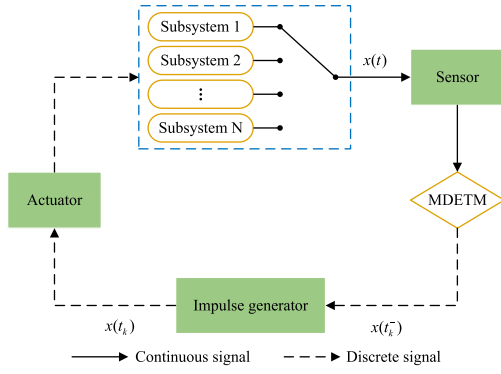


Fig. 1. Block diagram of ETIC for switched systems.

Definition 2 (Li & Yang, 2020). System (1) subject to (2) is globally weakly exponentially stable (GWES) if there exist constants $\kappa \geq 1$, $\rho > 0$, and functions $\varpi_1, \varpi_2 \in \mathcal{K}_\infty$ such that for $\xi \in \mathbb{P}\mathbb{C}_\gamma$, the solution $x(t)$ satisfies

$$\varpi_1(|x(t)|) \leq \kappa \varpi_2(\|\xi\|_\gamma) e^{-\rho(t-t_0)}, \quad \forall t \geq t_0.$$

Especially when $\varpi_1(s) = \varpi_2(s) = s^q$, $q \in \mathbb{R}_+$, the system is globally exponentially stable (GES).

Definition 3 (Chen & Zheng, 2009). Given locally Lipschitz functions $\mathcal{V}_i : \mathbb{R}^n \rightarrow \mathbb{R}_+$, $i \in I$, the upper right-hand Dini derivative of $\mathcal{V}_i$ along system (1) with (2) is

$$D^+ \mathcal{V}_i(\Gamma(0)) = \limsup_{\theta \rightarrow 0^+} \frac{1}{\theta} \left(\mathcal{V}_i(\Gamma(0) + \rho f_i(\Gamma)) - \mathcal{V}_i(\Gamma(0)) \right),$$

where $\Gamma \in \mathbb{P}\mathbb{C}_\gamma$.

3. Main results

This section provides sufficient conditions to rule out Zeno behavior and ensure exponential stability of the switched delay system under ETIC. Let $V_{\sigma(t)}(\cdot)$ denote the Lyapunov function associated with the subsystem activated by the switching signal $\sigma(t)$. The MDETM, which involves both the current mode $\sigma(t)$ and the mode $\sigma(t_{k-1})$ active at the previous triggering instant, is constructed as

$$t_k = \inf \{t \geq t_{k-1} : V_{\sigma(t)}(x(t)) - \Omega_{k-1}(t) \geq 0\}, \quad (3)$$

where

$$\Omega_{k-1}(t) = e^{\alpha_k - \rho t} (e^{\rho(t-t_{k-1})} V_{\sigma(t_{k-1})}(x(t_{k-1})) \vee e^{\rho t_0} V_0)$$

with $V_0 = \sup_{\theta \in [-\gamma, 0]} \{\max_{i \in I} V_i(x(t_0 + \theta))\}$, $\rho > 0$, and triggering parameters $\alpha_k \in \mathbb{R}_+$ satisfying

$$\sum_{k=1}^p \alpha_k \rightarrow \infty \text{ as } p \rightarrow \infty. \quad (4)$$

In practical digital implementation, the real-time monitoring of the triggering condition (3) is realized via discrete sampling approximation. A smaller sampling period improves triggering accuracy but increases the monitoring and computational burden, and vice versa. Therefore, the sampling period should be chosen as a trade-off between accuracy and implementation cost, with the key being that it must remain much smaller than the theoretical lower bound on the inter-event times. The following result establishes this bound and rules out Zeno behavior.

Theorem 1. Assume that there exist locally Lipschitz continuous functions $V_i : \mathbb{R}^n \rightarrow \mathbb{R}_+$ and positive constants $\lambda_1, \lambda_2, \alpha_k, \beta_k, \rho, \eta, \mu, N_0$ with $\alpha_k \in (0, \eta]$ and $\mu \geq 1$, such that for all $i, j \in I$ and $k \in \mathbb{Z}_+$

$$(H_1) \quad D^+ V_i(\Gamma(0)) \leq \lambda_1 V_i(\Gamma(0)) + \lambda_2 V_j(\Gamma(\theta)), \quad \forall \Gamma \in \mathbb{P}\mathbb{C}_\gamma, \quad \theta \in [-\gamma, 0];$$

$$(H_2) \quad V_i(x) \leq \mu V_j(x), \quad \forall x \in \mathbb{R}^n, \quad i \neq j;$$

$$(H_3) \quad V_i(g_k(x)) \leq e^{-\beta_k} V_j(x), \quad \forall x \in \mathbb{R}^n;$$

$$(H_4) \quad \inf_{k \in \mathbb{Z}_+} \{\alpha_k\} > N_0 \ln \mu;$$

$$(H_5) \quad \alpha_k + \sum_{m=1}^i (\alpha_{k-m} - \beta_{k-m}) \leq \eta, \quad \forall i \in \mathcal{L}_k, \quad \text{where } \mathcal{L}_k = \{1, 2, \dots, k-1\}, \quad k \geq 2.$$

Then system (1) subject to (2) over class $\mathcal{F}$ rules out Zeno behavior under MDETM (3). Furthermore,

$$t_k - t_{k-1} \geq \frac{\alpha_k - N_0 \ln \mu}{\lambda_2 e^{\eta + \rho \gamma} + \rho + \lambda_1 + \frac{\ln \mu}{\tau_a}}. \quad (5)$$

Proof. Define

$$\vartheta_{\sigma(t)}(t) = \begin{cases} e^{\rho(t-t_0)} V_{\sigma(t)}(x(t)), & t \in [t_0, \infty), \\ V_0, & t \in [t_0 - \gamma, t_0). \end{cases}$$

Then the MDETM (3) can be rewritten as

$$t_k = \inf \{t \geq t_{k-1} : \vartheta_{\sigma(t)}(t) \geq e^{\alpha_k} \omega(t_{k-1})\}, \quad (6)$$

where $\omega(t_{k-1}) = \vartheta_{\sigma(t_{k-1})}(t_{k-1}) \vee V_0$. Under the switching topology, two cases should be considered: whether switching occurs within the triggering interval $[t_{k-1}, t_k)$ or not.

Case I. Switching instants ℓ_ς exist within $[t_{k-1}, t_k)$ satisfying $t_{k-1} \leq \ell_1 < \ell_2 < \dots < \ell_M < t_k$, $M \in \mathbb{Z}_+$. Define

$$\varphi(t) = \mu^{N(t, t_{k-1})} e^{\lambda(t-t_{k-1})} \omega(t_{k-1}),$$

where $\lambda = \lambda_2 e^{\eta + \rho \gamma} + \rho + \lambda_1$. We shall show

$$\vartheta_{\sigma(t)}(t) \leq \varphi(t), \quad t \in [t_{k-1}, t_k). \quad (7)$$

(i) Firstly, we prove that (7) holds for $t \in [t_{k-1}, \ell_1)$, i.e.,

$$\vartheta_{\sigma(t_{k-1})}(t) \leq e^{\lambda(t-t_{k-1})} \omega(t_{k-1}), \quad t \in [t_{k-1}, \ell_1). \quad (8)$$

This holds trivially for $t = t_{k-1}$. Suppose there exists $t \in (t_{k-1}, \ell_1)$ such that $\vartheta_{\sigma(t_{k-1})}(t) > \varphi(t)$. Let $t^* = \inf \{t \in (t_{k-1}, \ell_1) : \vartheta_{\sigma(t_{k-1})}(t) > \varphi(t)\}$. Then $\vartheta_{\sigma(t_{k-1})}(t^*) = \varphi(t^*)$, and

$$\vartheta_{\sigma(t_{k-1})}(t) > \varphi(t), \quad t \in (t^*, t^* + \varepsilon^*), \quad (9)$$

where ε^* denotes a sufficiently small positive number. Noticing that for $s \in [t_0 - \gamma, t_0)$,

$$\vartheta_{\sigma(s)}(s) = V_0.$$

For $s \in [t_0, t_1)$, we obtain

$$\vartheta_{\sigma(s)}(s) \leq \vartheta_{\sigma(t_1^-)}(t_1^-) = e^{\alpha_1} \omega(t_0) \leq e^{\alpha_1} V_0.$$

If a switching instant coincides with the triggering instant t_1 , without loss of generality, one may take $i = \sigma(t_1)$ and $j = \sigma(t_1^-)$ with $i \neq j$ in condition (H₃). Then it follows that

$$\vartheta_{\sigma(t_1)}(t_1) \leq e^{-\beta_1} \vartheta_{\sigma(t_1^-)}(t_1^-) \leq e^{\alpha_1 - \beta_1} V_0.$$

In particular, if no such switching occurs at t_1 , then $i = \sigma(t_1) = \sigma(t_1^-) = j$, and the above inequality still holds. For $s \in [t_1, t_2)$, it yields

$$\vartheta_{\sigma(s)}(s) \leq \vartheta_{\sigma(t_2^-)}(t_2^-) = e^{\alpha_2} \omega(t_1) \leq (e^{\alpha_2 + \alpha_1 - \beta_1} \vee e^{\alpha_2}) V_0.$$

For each $k \in \mathbb{Z}_+$, define

$$\Xi_k = \begin{cases} \alpha_1, & k = 1, \\ \max_{i \in \mathcal{L}_k} \left\{ \alpha_k + \sum_{m=1}^i (\alpha_{k-m} - \beta_{k-m}) \right\}, & k \geq 2. \end{cases}$$

By recursively applying the above derivation and combining it with (H₅), we deduce

$$\vartheta_{\sigma(s)}(s) \leq \begin{cases} \vartheta_{\sigma(t_k^-)}(t_k^-), & s \in [t_{k-1}, t_k), \\ \vartheta_{\sigma(t_{k-1}^-)}(t_{k-1}^-), & s \in [t_{k-2}, t_{k-1}), \\ \vdots \\ \vartheta_{\sigma(t_1^-)}(t_1^-), & s \in [t_0, t_1), \\ V_0, & s \in [t_0 - \gamma, t_0) \end{cases}$$

$$\leq \begin{cases} (e^{\bar{\alpha}_k} \vee e^{\alpha_k})V_0, & s \in [t_{k-1}, t_k), \\ (e^{\bar{\alpha}_{k-1}} \vee e^{\alpha_{k-1}})V_0, & s \in [t_{k-2}, t_{k-1}), \\ \vdots \\ e^{\alpha_1}V_0, & s \in [t_0, t_1), \\ V_0, & s \in [t_0 - \gamma, t_0) \end{cases} \leq e^\eta V_0. \quad (10)$$

Since $V_0 \leq \omega(t_{k-1}) \leq \varphi(t) \leq \vartheta_{\sigma(t_{k-1})}(t)$ for $t \in [t^*, t^* + \varepsilon^*)$, it follows

$$\vartheta_{\sigma(s)}(s) \leq e^\eta \vartheta_{\sigma(t_{k-1})}(t), \quad s \in [t_0 - \gamma, t_k). \quad (11)$$

For $\theta \in [-\gamma, 0]$, condition (H₁) yields

$$\begin{aligned} D^+ \vartheta_{\sigma(t_{k-1})}(t) &= e^{\rho(t-t_0)}(\rho V_{\sigma(t_{k-1})}(x(t)) + D^+ V_{\sigma(t_{k-1})}(x(t))) \\ &\leq (\rho + \lambda_1) \vartheta_{\sigma(t_{k-1})}(t) + \lambda_2 e^{-\rho\theta} \vartheta_{\sigma(t+\theta)}(t + \theta) \\ &\leq \lambda \vartheta_{\sigma(t_{k-1})}(t), \quad t \in [t^*, t^* + \varepsilon^*). \end{aligned} \quad (12)$$

For $\delta^* \in (0, \varepsilon^*)$, we obtain

$$\vartheta_{\sigma(t_{k-1})}(t^* + \delta^*) \leq e^{\lambda(t^* + \delta^* - t^*)} \vartheta_{\sigma(t_{k-1})}(t^*) = \varphi(t^* + \delta^*),$$

which contradicts (9). Thus, inequality (8) holds.

(ii) Suppose that inequality (7) is true for $t \in [t_{k-1}, \ell_\zeta]$, where $\zeta \in \{1, 2, \dots, M-1\}$. We proceed to verify its validity for $t \in [\ell_\zeta, \ell_{\zeta+1})$, i.e., $\vartheta_{\sigma(\ell_\zeta)}(t) \leq \mu^\zeta e^{\lambda(t-t_{k-1})} \omega(t_{k-1})$, $t \in [\ell_\zeta, \ell_{\zeta+1})$. (13)

For $t = \ell_\zeta$, condition (H₂) implies that

$$\vartheta_{\sigma(\ell_\zeta)}(\ell_\zeta) \leq \mu \vartheta_{\sigma(\ell_\zeta^-)}(\ell_\zeta^-) \leq \mu \mu^{\zeta-1} e^{\lambda(\ell_\zeta - t_{k-1})} \omega(t_{k-1}) = \varphi(\ell_\zeta).$$

If (13) is not satisfied, there exists $t^\circ = \inf\{t \in (\ell_\zeta, \ell_{\zeta+1}) : \vartheta_{\sigma(\ell_\zeta)}(t) > \varphi(t)\}$ such that $\vartheta_{\sigma(\ell_\zeta)}(t^\circ) = \varphi(t^\circ)$, and $\vartheta_{\sigma(\ell_\zeta)}(t) > \varphi(t)$, $t \in (t^\circ, t^\circ + \varepsilon^\circ)$, where ε° denotes a sufficiently small positive number. Following the same reasoning as in (i), the validity of inequality (13) can be verified.

(iii) For $t \in [\ell_M, t_k]$, analogous to the analysis in (ii), we conclude that

$$\vartheta_{\sigma(\ell_M)}(t) \leq \mu^M e^{\lambda(t-t_{k-1})} \omega(t_{k-1}), \quad t \in [\ell_M, t_k].$$

The preceding analysis confirms the validity of (7). By Definition 1, we have

$$\vartheta_{\sigma(t)}(t) \leq e^{N_0 \ln \mu + (\lambda + \frac{\ln \mu}{\tau_a})(t-t_{k-1})} \omega(t_{k-1}), \quad t \in [t_{k-1}, t_k].$$

Combining with $\vartheta_{\sigma(t_k^-)}(t_k^-) = e^{\alpha_k} \omega(t_{k-1})$, we obtain

$$e^{\alpha_k} \omega(t_{k-1}) \leq e^{N_0 \ln \mu + (\lambda + \frac{\ln \mu}{\tau_a})(t_k - t_{k-1})} \omega(t_{k-1}),$$

and hence

$$t_k - t_{k-1} \geq \frac{\alpha_k - N_0 \ln \mu}{\lambda + \frac{\ln \mu}{\tau_a}}.$$

Case II. No switching instants exist within $[t_{k-1}, t_k]$. Since $N(t, t_{k-1}) = 0$, the derivation follows directly from the analytical approach in (i) of Case I. Hence,

$$\vartheta_{\sigma(t)}(t) \leq e^{\lambda(t-t_{k-1})} \omega(t_{k-1}), \quad t \in [t_{k-1}, t_k].$$

Together with $\vartheta_{\sigma(t_k^-)}(t_k^-) = e^{\alpha_k} \omega(t_{k-1})$, it yields

$$t_k - t_{k-1} \geq \frac{\alpha_k}{\lambda}.$$

Hence, regardless of whether switching occurs within the triggering interval, it follows from (4) and (H₄) that

$$t_k = \sum_{h=1}^k (t_h - t_{h-1}) + t_0 \rightarrow \infty \text{ as } k \rightarrow \infty,$$

which rules out Zeno behavior. Furthermore, the triggered impulse sequence $\{t_k\}$ satisfies

$$t_k - t_{k-1} \geq \frac{\alpha_k - N_0 \ln \mu}{\lambda_2 e^{\eta + \rho\gamma} + \rho + \lambda_1 + \frac{\ln \mu}{\tau_a}}.$$

This completes the proof. $\square$

Remark 1. It follows from the lower bound on the inter-event times in (5) that a smaller triggering parameter α_k or a larger decay rate ρ tends to shorten the inter-event times and may increase the impulse frequency, whereas a larger α_k or a smaller ρ tends to have the opposite effect. Unlike existing results that establish a uniform positive lower bound for all triggering intervals (Fei et al., 2020; Ge et al., 2024; Liu et al., 2018; Qi et al., 2020), the lower bound derived in this paper explicitly depends on α_k , and thus may vary with k . This k -dependence enables the admissible lower bound of each triggering interval to be adjusted individually, offering greater design flexibility. As a special case, when $\alpha_k = \beta_k = \zeta$ for all $k \in \mathbb{Z}_+$, this bound reduces to a constant, as shown in Corollary 1 below.

Corollary 1. Under conditions (H₁)–(H₃), assume that $\alpha_k = \beta_k = \zeta$ with $\zeta > N_0 \ln \mu$ for all $k \in \mathbb{Z}_+$. Then system (1) subject to (2) over class $\mathcal{F}$ rules out Zeno behavior under MDETM (3). Furthermore,

$$t_k - t_{k-1} \geq \frac{\zeta - N_0 \ln \mu}{\lambda_2 e^{\zeta + \rho\gamma} + \rho + \lambda_1 + \frac{\ln \mu}{\tau_a}}.$$

Next, sufficient criteria for exponential stability are derived.

Theorem 2. Under conditions in Theorem 1, assume that there exist functions $\varpi_1, \varpi_2 \in \mathcal{K}_\infty$, such that for all $i \in I$

$$(H_6) \quad \varpi_1(|x|) \leq V_i(x) \leq \varpi_2(|x|), \quad \forall x \in \mathbb{R}^n.$$

Then system (1) subject to (2) over class $\mathcal{F}$ is GWES under MDETM (3).

Proof. From the derivation of (10), we obtain

$$\vartheta_{\sigma(t)}(t) \leq \vartheta_{\sigma(t_k^-)}(t_k^-) \leq (e^{\bar{\alpha}_k} \vee e^{\alpha_k})V_0, \quad \forall t \in [t_{k-1}, t_k].$$

Combining with (H₅), it yields

$$e^{\rho(t-t_0)} V_{\sigma(t)}(x(t)) = \vartheta_{\sigma(t)}(t) \leq e^\eta V_0, \quad \forall t \geq t_0.$$

By condition (H₆), we derive

$$\varpi_1(|x(t)|) \leq e^\eta \varpi_2(\|x\|_\gamma) e^{-\rho(t-t_0)}, \quad \forall t \geq t_0,$$

thereby confirming the GWES of the system. This completes the proof. $\square$

Corollary 2. Under conditions in Theorem 1, assume that there exist positive constants c_1, c_2 , and q , such that for all $i \in I$

$$(H'_6) \quad c_1 |x|^q \leq V_i(x) \leq c_2 |x|^q, \quad \forall x \in \mathbb{R}^n.$$

Then system (1) subject to (2) over class $\mathcal{F}$ is GES under MDETM (3).

Remark 2. The continuous evolution of the system state is governed by condition (H₁), where λ_1 and λ_2 characterize the delay-independent and delay-dependent dynamical terms, respectively. Condition (H₂) describes the variation of the Lyapunov functions at switching instants, with the parameter μ reflecting the possible amplification induced by subsystem switching. Condition (H₃) specifies the impulse jump at triggering instants through the impulse strengths β_k , and therefore can be interpreted as an impulse generator. Conditions (H₄) and (H₅) jointly provide guidelines for selecting the triggering parameters α_k . The former reflects the influence of the switching topology on the choice of α_k , while the latter establishes a coupling relation between the triggering parameters α_k and the impulse strengths β_k . Notably, condition (H₅) imposes a boundedness constraint on the cumulative effect of the triggering parameters and the impulse strengths, rather than on their individual magnitudes. It reveals that, under the stabilizing effect of impulsive control, the cumulative influence of the triggering parameters can be counterbalanced by the impulse strengths. Consequently, the cumulative term $\sum_{m=1}^i (\alpha_{k-m} - \beta_{k-m})$ in (H₅) is not required to grow monotonically and can remain bounded, thus avoiding overly restrictive conditions on the triggering parameters. In fact, with appropriately chosen impulse strengths, this cumulative term may remain constant (as in Corollary 1) or even exhibit a decreasing trend (as shown in Corollary 3 below), further demonstrating the flexibility of the proposed framework.

Corollary 3. Under conditions (H₁)-(H₄) and (H₆), assume that $\alpha_{2k-1} - \beta_{2k-1} = -\zeta_1$, $\alpha_{2k} - \beta_{2k} = \zeta_2$, and $\alpha_k \in (0, \eta - \zeta_2]$ for all $k \in \mathbb{Z}_+$, where $\zeta_1, \zeta_2 > 0$ satisfy $\zeta_2 < \min\{\zeta_1, \frac{\eta}{2}\}$. Then system (1) subject to (2) over class $\mathcal{F}$ is GWES under MDETM (3).

Remark 3. In time-triggered impulsive control strategies, constraints are typically imposed on the rate coefficients of the continuous dynamics to restrict the divergence rate, while an upper bound is imposed on the impulsive intervals to guarantee that impulses occur sufficiently frequently (Kang et al., 2023; Liu et al., 2011; Ren & Xiong, 2019; Wang et al., 2022). In contrast, the present results impose no restrictions on the coefficients λ_1 and λ_2 associated with the continuous dynamics, and the impulse instants are fully generated by the MDETM. In this sense, the requirements on both the continuous dynamics and impulse actions are significantly relaxed compared with the maximum or average impulsive interval conditions established in the aforementioned literature. In fact, within the ETIC framework, the values of the rate coefficients solely influence the lower bound on the inter-event times. Specifically, larger rate coefficients imply higher impulse frequency, whereas smaller ones indicate lower impulse frequency. This implicit relationship can be inferred from (5).

Remark 4. In the context of Lyapunov stability, various ETIC schemes have been developed for nonlinear systems, including those imposing an explicit upper bound on the inter-event times (Zhu et al., 2018), incorporating a forced impulse sequence (Li et al., 2020), and adopting a dynamic ETM with a prescribed minimum sampling interval (Ai et al., 2022). However, these approaches inevitably introduce additional restrictions on the impulse sequence and, more importantly, neglect the effects of time delay and switching topology, making them inapplicable to switched delay systems. To address these issues, this paper develops an MDETM that operates without preassigned restrictions, thereby offering greater flexibility in impulse generation. By incorporating both the exponential attenuation term $e^{-\rho(t-t_0)}V_0$ and the switching signal $\sigma(t)$ into the triggering threshold, the MDETM simultaneously accounts for the coupling effects of time delay and switching topology. It should be emphasized that the term $e^{-\rho(t-t_0)}V_0$ is not introduced to artificially enlarge the triggering threshold or reduce the impulse frequency; rather, as shown in the proof of Theorem 1, it leads to the auxiliary expression $e^{\rho(t_{k-1}-t_0)}V_{\sigma(t_{k-1})}(x(t_{k-1})) \vee V_0$, which effectively connects the delayed historical state with the current state. In this way, this term becomes essential for addressing delayed state information within the Lyapunov-Razumikhin framework.

Remark 5. To date, substantial progress has been made in stabilizing switched systems using ETC (Li et al., 2018; Xiao et al., 2019; Zhao et al., 2022; Zobiri et al., 2020). However, these studies rely on a common requirement that the control input must be updated at least once within each switching interval. Although ETIC strategies developed in Gao et al. (2025), Zhang et al. (2024, 2022) have successfully removed this restriction, the corresponding results become invalid when delay is involved. In contrast, this paper effectively addresses the adverse effects of time delay and establishes exponential stability for switched delay systems within the proposed ETIC framework.

4. Synchronization of SNNs

This section extends the proposed ETIC strategy to achieve synchronization of SNNs.

Consider the SNNs

$$\begin{cases} \dot{x}(t) = -E_{\sigma(t)}x(t) + A_{\sigma(t)}\phi(x(t)) + B_{\sigma(t)}\phi(x(t-\gamma(t))) + J, & t \geq t_0, \\ x_{t_0} = \psi_1, \end{cases} \quad (14)$$

where $x(t) = (x_1(t), \dots, x_n(t))^T \in \mathbb{R}^n$ denotes the neuron state, $\phi(x(t)) = (\phi_1(x_1(t)), \dots, \phi_n(x_n(t)))^T \in \mathbb{R}^n$ is the neuron activation function, $J \in \mathbb{R}^n$ represents an external input, the time delay $\gamma(t)$ satisfies $0 \leq \gamma(t) \leq \gamma$, and

the switching signal $\sigma(t) : [t_0, \infty) \rightarrow \mathcal{I} = \{1, 2, \dots, N\}$. $E_{\sigma(t)} > 0$ is a diagonal matrix, $A_{\sigma(t)}$ and $B_{\sigma(t)}$ represent the connection weight matrices. $\psi_1 \in \mathbb{P}\mathbb{C}_\gamma$ is the initial function.

System (14) serves as the drive system, and the corresponding response system is

$$\begin{cases} \dot{y}(t) = -E_{\sigma(t)}y(t) + A_{\sigma(t)}\phi(y(t)) + B_{\sigma(t)}\phi(y(t-\gamma(t))) + J + u(t), & t \geq t_0, \\ y_{t_0} = \psi_2, \end{cases} \quad (15)$$

where $\psi_2 \in \mathbb{P}\mathbb{C}_\gamma$, and $u(t)$ is the control input defined as

$$u(t) = \sum_{k \in \mathbb{Z}_+} D_k z(t) \delta(t - t_k)$$

with synchronization error $z(t) = y(t) - x(t)$, the Dirac delta function $\delta(\cdot)$, the impulse sequence $\{t_k\}$, and the impulsive control gain D_k .

Then the error system is

$$\begin{cases} \dot{z}(t) = -E_{\sigma(t)}z(t) + A_{\sigma(t)}\hbar(z(t)) + B_{\sigma(t)}\hbar(z(t-\gamma(t))), & t \neq t_k, \quad t \geq t_0, \\ z(t) = (I + D_k)z(t^-), & t = t_k, \quad k \in \mathbb{Z}_+, \\ z_{t_0} = \xi, \end{cases} \quad (16)$$

where $\hbar(z(\cdot)) = \phi(y(\cdot)) - \phi(x(\cdot))$, and $\xi = \psi_2 - \psi_1$.

Definition 4 (Yang et al., 2011). The drive-response systems (14) and (15) are globally exponentially synchronized if the error system (16) is GES.

Assumption 1. For each $m \in \Lambda$, the neuron activation function $\phi_m(\cdot)$ satisfies

$$w_m^- \leq \frac{\phi_m(v_1) - \phi_m(v_2)}{v_1 - v_2} \leq w_m^+, \quad \forall v_1, v_2 \in \mathbb{R}, \quad v_1 \neq v_2,$$

where w_m^- and w_m^+ are known constants.

For later use, we define $W_1 = \text{diag}\{w_1^- w_1^+, \dots, w_n^- w_n^+\}$ and $W_2 = \text{diag}\left\{\frac{w_1^- + w_1^+}{2}, \dots, \frac{w_n^- + w_n^+}{2}\right\}$. By selecting appropriate multiple Lyapunov functions, the following LMI-based synchronization criteria for the drive-response systems (14) and (15) can be directly obtained from Corollary 2.

Theorem 3. Under Assumption 1 and conditions (H₄)-(H₅), assume that there exist matrices $P_i > 0$, diagonal matrices $R_i > 0$, $S_i > 0$, real matrices U_k with appropriate dimensions, positive constants $\lambda_1, \lambda_2, \alpha_k, \beta_k, \rho, \eta, \mu, N_0$ with $\alpha_k \in (0, \eta]$ and $\mu \geq 1$, such that for all $i, j \in \mathcal{I}$ and $k \in \mathbb{Z}_+$

$$\Psi_{ij} = \begin{pmatrix} \Phi_i & 0 & \Theta_i & P_i B_i \\ \star & \Upsilon_{ij} & 0 & W_2 S_i \\ \star & \star & -R_i & 0 \\ \star & \star & \star & -S_i \end{pmatrix} \leq 0, \quad (17)$$

$$\begin{pmatrix} -e^{-\beta_k} P_j & U_k^T \\ \star & -P_i \end{pmatrix} \leq 0, \quad (18)$$

$$P_i \leq \mu P_j, \quad i \neq j, \quad (19)$$

where

$$\Phi_i = -P_i E_i - E_i^T P_i - W_1 R_i - \lambda_1 P_i,$$

$$\Theta_i = P_i A_i + W_2 R_i,$$

$$\Upsilon_{ij} = -W_1 S_i - \lambda_2 P_j.$$

Then the drive-response systems (14) and (15) over class $\mathcal{F}$ are globally exponentially synchronized under the control gain $D_k = P_i^{-1} U_k - I$ and the following MDETM

$$t_k = \inf \{t \geq t_{k-1} : z^T(t) P_{\sigma(t)} z(t) - \Omega_{k-1}(t) \geq 0\}, \quad (20)$$

where

$$\Omega_{k-1}(t) = e^{\alpha_k - \rho t} (e^{\rho t_{k-1}} z^T(t_{k-1}) P_{\sigma(t_{k-1})} z(t_{k-1}) \vee e^{\rho t_0} V_0)$$

with $V_0 = \sup_{\theta \in [-\gamma, 0]} \{\max_{i \in \mathcal{I}} z^T(t_0 + \theta) P_i z(t_0 + \theta)\}$.

Algorithm 1 ETIC for synchronization of SNNs.**Require:**

- 1: Initialize the parameters α_k , ρ , and matrices P_i , D_k ;
- 2: Input the initial functions ψ_1 , ψ_2 , and initial time t_0 ;

Ensure:

- 3: **for** $t \in [t_0, t_{\text{end}}]$ **do**
- 4: Update the switching signal $\sigma(t)$ and states $x(t)$, $y(t)$;
- 5: Calculate the error state $z(t) \leftarrow y(t) - x(t)$;
- 6: Calculate the functions $V_{\sigma(t)}(z(t))$ and $\Omega_{k-1}(t)$;
- 7: **if** $V_{\sigma(t)}(z(t)) \geq \Omega_{k-1}(t)$ **then**
- 8: Update the triggering instant $t_k \leftarrow t$;
- 9: Use impulsive control $y(t_k) \leftarrow y(t_k^-) + D_k z(t_k^-)$;
- 10: Record $\sigma(t_k)$ and update $k \leftarrow k + 1$;
- 11: **end if**
- 12: **end for**

Proof. Define $V_i(t) := V_i(z(t)) = z^T(t)P_i z(t)$, $i \in \mathcal{I}$. Then

$$D^+ V_i(t) = z^T(t)(-P_i E_i - E_i^T P_i)z(t) + 2z^T(t)P_i A_i \hat{h}(z(t)) + 2z^T(t)P_i B_i \hat{h}(z(t - \gamma(t))). \quad (21)$$

For any $n \times n$ diagonal matrices $R_i > 0$ and $S_i > 0$, Assumption 1 implies that

$$\chi_1^T(t) \begin{pmatrix} W_1 R_i & -W_2 R_i \\ \star & R_i \end{pmatrix} \chi_1(t) \leq 0, \quad (22)$$

and

$$\chi_1^T(t - \gamma(t)) \begin{pmatrix} W_1 S_i & -W_2 S_i \\ \star & S_i \end{pmatrix} \chi_1(t - \gamma(t)) \leq 0, \quad (23)$$

where $\chi_1(\cdot) = (z^T(\cdot), \hat{h}^T(z(\cdot)))^T$. Substituting (22) and (23) into (21), one has

$$\begin{aligned} D^+ V_i(t) &\leq z^T(t)(-P_i E_i - E_i^T P_i - W_1 R_i)z(t) \\ &\quad + 2z^T(t)(P_i A_i + W_2 R_i)\hat{h}(z(t)) + 2z^T(t)P_i B_i \hat{h}(z(t - \gamma(t))) \\ &\quad - z^T(t - \gamma(t))W_1 S_i z(t - \gamma(t)) + 2z^T(t - \gamma(t))W_2 S_i \hat{h}(z(t - \gamma(t))) \\ &\quad - \hat{h}^T(z(t))R_i \hat{h}(z(t)) - \hat{h}^T(z(t - \gamma(t)))S_i \hat{h}(z(t - \gamma(t))) \\ &= \chi_2^T(t) \Psi_{ij} \chi_2(t) + \lambda_1 V_i(t) + \lambda_2 V_j(t - \gamma(t)), \end{aligned}$$

where $\chi_2(t) = (z^T(t), z^T(t - \gamma(t)), \hat{h}^T(z(t)), \hat{h}^T(z(t - \gamma(t))))^T$. Combining with (17), we get

$$D^+ V_i(t) \leq \lambda_1 V_i(t) + \lambda_2 V_j(t - \gamma(t)). \quad (24)$$

Using the Schur complement lemma on (18) gives

$$(I + D_k)^T P_i (I + D_k) \leq e^{-\beta_k} P_j,$$

which further yields

$$V_i(t_k) \leq e^{-\beta_k} V_j(t_k^-). \quad (25)$$

Furthermore, it follows from (19) that

$$V_i(\ell_\zeta^-) = z^T(\ell_\zeta^-)P_i z(\ell_\zeta^-) \leq \mu V_j(\ell_\zeta^-). \quad (26)$$

In view of (24)-(26), all conditions in Corollary 2 hold, thereby confirming the GES of the error system (16). This completes the proof. $\square$

Remark 6. Theorem 3 presents LMI-based criteria for synchronization of SNNs under ETIC. The design philosophy is to convert the conditions in Corollary 2 into tractable matrix inequalities. Specifically, inequality (17) corresponds to the continuous dynamics condition (H₁), whereas inequalities (18) and (19) follow from the discrete dynamics conditions (H₃) and (H₂), which characterize the impulse actions and switching behavior, respectively. The LMIs are formulated for all possible pairs of subsystems $(i, j) \in \mathcal{I} \times \mathcal{I}$, which causes the number of LMI conditions to grow quadratically with the number of subsystems N . This universal formulation may introduce additional computational burden, especially

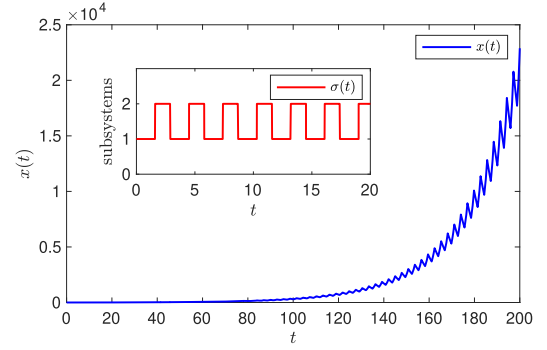


Fig. 2. Switching signal and state trajectory of system (1).

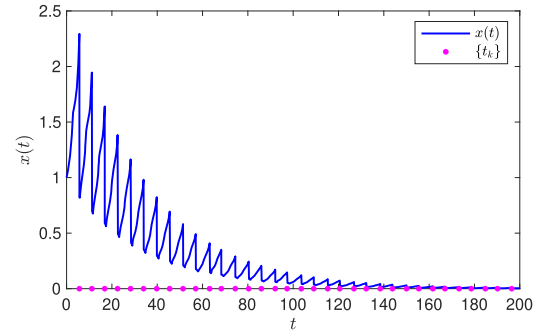


Fig. 3. Triggering instants and state trajectory of system (1) subject to (2) under MDET (27) ($d_k = -0.64$, $\zeta = 1$).

when N is large or the switching topology is sparse, as the conditions must hold for all subsystem pairs including those that may never actually occur. Nevertheless, it has its own rationale. In principle, if the admissible subsystem pairs are known a priori and can be characterized by a set $\mathcal{E} \subseteq \mathcal{I} \times \mathcal{I}$, one may impose the LMIs only on $(i, j) \in \mathcal{E}$, thereby reducing both conservatism and computational cost. However, for the time-varying delay considered in this paper, where no additional restrictions are imposed on the switching behavior within the delay interval, it is generally difficult to determine in advance which subsystem mode is active at each historical instant. Hence, the present formulation over all subsystem pairs is adopted to provide a general and verifiable sufficient condition for switched delay systems, despite the potential conservatism it may entail. To implement Theorem 3, the co-design procedure for the control gain and triggering parameters is outlined as follows.

- Step 1.** Initialize the system parameters, including the matrices E_i , A_i , B_i , W_1 , and W_2 .
- Step 2.** Given the scalars λ_1 , λ_2 , β_k , and μ , solve the LMI feasibility problem subject to the constraints (17)-(19) to determine the matrices P_i , R_i , S_i , and U_k . Then calculate the control gain D_k .
- Step 3.** Identify the ADT constant τ_a and the chatter bound N_0 over class $\mathcal{F}$.
- Step 4.** Select the parameters α_k and η to satisfy conditions (H₄) and (H₅).

Moreover, Algorithm 1 details the execution procedure for synchronization of SNNs under the proposed ETIC strategy.

5. Examples

This section provides two examples to confirm the feasibility of the developed approach.

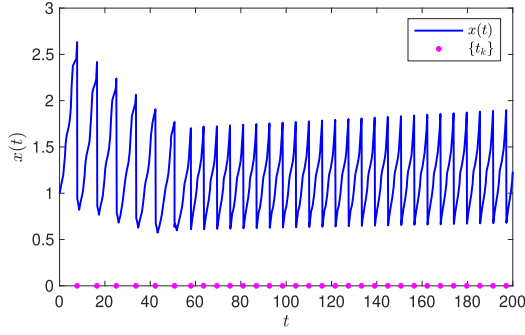


Fig. 4. Triggering instants and state trajectory of system (1) subject to (2) under MDETM (27) ($d_k = -0.64$, $\zeta = 1.2$).

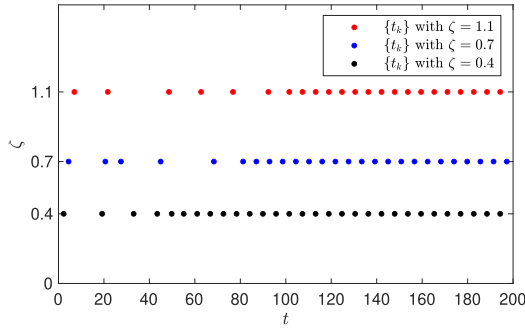


Fig. 5. Triggering instants of system (1) subject to (2) under MDETM (27) with different ζ ($d_k = -0.9$).

Example 1. Consider system (1) with $f_1(x_t) = -0.1x(t) + 0.3 \tanh(x(t - \gamma(t)))$ and $f_2(x_t) = 0.4x(t) - 0.2x(t - \gamma(t))$, where $\gamma(t) = 1 - 0.5e^{-0.2t}$. The switching sequence is $\ell_{2\zeta-1} = 2.9\zeta - 1.3$, $\ell_{2\zeta} = 2.9\zeta$, $\zeta \in \mathbb{Z}_+$, with an average dwell time $\tau_a = 1.45$. The initial function is $\xi = \cos t$, $t \in [-1, 0]$. Fig. 2 shows that system (1) is unstable. Next, consider system (1) under impulses (2) with $x(t_k) = (1 + d_k)x(t_k^-)$, where d_k denotes the impulsive control gain. By Corollary 2, we select $V(x(t)) = |x(t)|$. Setting $\alpha_k = \beta_k = \zeta$, conditions (H₁), (H₂), (H₄), and (H₅) are satisfied with $\lambda_1 = 0.4$, $\lambda_2 = 0.3$, $\mu = 1$, $N_0 = 2$, and $\eta = \zeta$. Combined with condition (H₃), the system achieves GES under $d_k \in [-e^{-\zeta} - 1, e^{-\zeta} - 1]$ and the MDETM

$$t_k = \inf \{t \geq t_{k-1} : |x(t)| - \Omega_{k-1}(t) \geq 0\}, \quad (27)$$

where

$$\Omega_{k-1}(t) = e^{\zeta - 0.03t} (e^{0.03t_{k-1}} |x(t_{k-1})| \vee V_0)$$

with $V_0 = \sup_{\theta \in [-1, 0]} \{|\cos \theta|\}$. For $\zeta = 1$, the corresponding control gain is specified as $d_k \in [-1.3679, -0.6321]$. With $d_k = -0.64$, the triggering instants and state trajectory of system (1) subject to (2) are given in Fig. 3, confirming that the system achieves GES under MDETM (27). If the triggering parameter is slightly changed to $\zeta = 1.2$, the stability criteria in Corollary 2 are violated ($d_k = -0.64 \notin [-1.3012, -0.6988]$). In this situation, the system cannot be stabilized, see Fig. 4. Moreover, Fig. 5 compares the triggering instants of MDETM (27) for different triggering parameters ($\zeta = 0.4, 0.7, 1.1$) with $d_k = -0.9$. Notably, a larger triggering parameter leads to a lower triggering frequency.

Example 2. Consider SNNs (14) with

$$E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 2 & 10 \\ 0.1 & 1.8 \end{pmatrix}, \quad B_1 = \begin{pmatrix} -1.6 & 0.2 \\ 0.1 & -1.5 \end{pmatrix},$$

$$E_2 = \begin{pmatrix} 1.2 & 0 \\ 0 & 1.1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1.7 & 12 \\ 0.1 & 1.9 \end{pmatrix}, \quad B_2 = \begin{pmatrix} -1 & 0 \\ 0.12 & -1.6 \end{pmatrix},$$

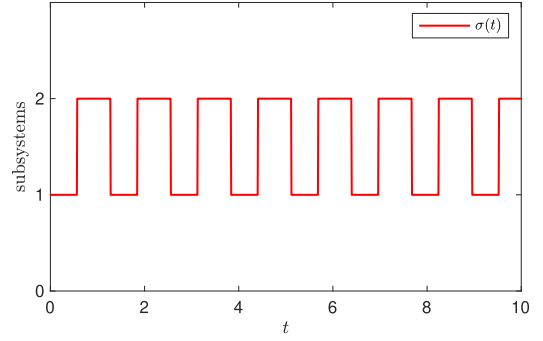


Fig. 6. Switching signal.

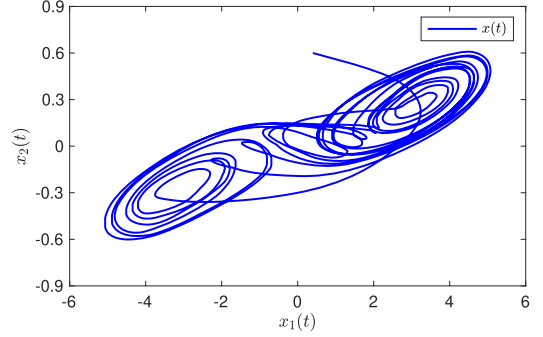


Fig. 7. Chaotic behavior of drive system (14).

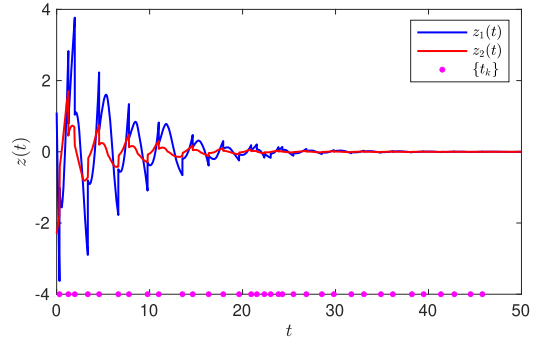


Fig. 8. Triggering instants and state trajectories of error system (16) under MDETM (20).

$\phi(s) = \tanh(s)$, $\gamma(t) = 0.85 + 0.15|\sin t|$, and $J = (0, 0)^T$. The switching sequence in Fig. 6 is defined by $\ell_{2\zeta-1} = 1.28\zeta - 0.71$, $\ell_{2\zeta} = 1.28\zeta$, $\zeta \in \mathbb{Z}_+$, yielding an average dwell time $\tau_a = 0.64$. Given the initial function $\psi_1 = (0.4, 0.6)^T$, system (14) exhibits chaotic behavior, as illustrated in Fig. 7. Let $\lambda_1 = 14$, $\lambda_2 = 3.2$, $\alpha_k = 0.32$, $\beta_k = 0.32$, $\rho = 0.3$, $\eta = 0.32$, $\mu = 1.1$, and $N_0 = 2$. Assumption 1 is satisfied with $W_1 = \text{diag}\{0, 0\}$ and $W_2 = \text{diag}\{0.5, 0.5\}$. The feasible solutions derived from Theorem 3 are

$$P_1 = \begin{pmatrix} 3.3155 & 0.2468 \\ 0.2468 & 12.4123 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 3.2287 & 0.2710 \\ 0.2710 & 12.5405 \end{pmatrix},$$

$$R_1 = \begin{pmatrix} 63.7905 & 0 \\ 0 & 63.7905 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 86.7813 & 0 \\ 0 & 86.7813 \end{pmatrix},$$

$$S_1 = \begin{pmatrix} 23.0916 & 0 \\ 0 & 23.0916 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 26.1035 & 0 \\ 0 & 26.1035 \end{pmatrix},$$

and the control gain is specified as

$$D_k = \begin{pmatrix} -0.7243 & 0.0015 \\ 0.0002 & -0.7357 \end{pmatrix}.$$

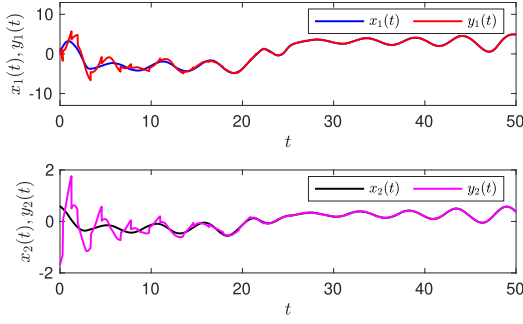


Fig. 9. State trajectories of drive-response systems (14) and (15) under MDETM (20).

Table 1

Comparisons of different triggering mechanisms.

Mechanism	N_I	T_I	J
Proposed MDETM (20)	34	1.3470	19.4181
Method in (Tian & Lu, 2026)	69	0.7200	19.4889
Method in (Gao et al., 2020)	78	0.6400	19.4265

Algorithm 2 The diffusion process of image encryption.

Require:

1: The chaotic sequences $C_l(s)$ and pixel sequences $\mathcal{O}_l(s)$

Ensure:

2: **for** $l = 1$ to 3 **do**

3: $\mathcal{T}_l(1) \leftarrow \mathcal{O}_l(1) \oplus ((C_l(pq) + \mathcal{O}_l(pq)) \bmod 256)$;

4: **for** $s = 2$ to pq **do**

5: $\mathcal{T}_l(s) \leftarrow \mathcal{O}_l(s) \oplus ((C_l(s-1) + \mathcal{T}_l(s-1)) \bmod 256)$;

6: **end for**

7: $\mathcal{N}_l(pq) \leftarrow \mathcal{T}_l(pq) \oplus ((C_l(1) + \mathcal{T}_l(1)) \bmod 256)$;

8: **for** $s = pq - 1$ to 1 **do**

9: $\mathcal{N}_l(s) \leftarrow \mathcal{T}_l(s) \oplus ((C_l(s+1) + \mathcal{N}_l(s+1)) \bmod 256)$;

10: **end for**

11: **end for**

The initial function of response system (15) is $\psi_2 = (1.5, -1.7)^T$. Under MDETM (20), Fig. 8 depicts the GES of error system (16) and the triggering instants. Fig. 9 further shows the synchronization between systems (14) and (15), demonstrating the applicability of the proposed method in addressing time-varying delay effects. Moreover, to further demonstrate the advantage of the proposed ETIC strategy, we compare it with two representative time-triggered impulsive schemes (Gao et al., 2020; Tian & Lu, 2026). To ensure a fair comparison, all simulations are conducted under the same initial conditions and with the same impulsive control gain. In addition, the performance cost index $J = \int_0^{t_{\text{end}}} z^T(t)z(t)dt$ is introduced to quantify the accumulated synchronization error (Xu et al., 2008). For the asynchronous impulse-switching sequence based on Tian and Lu (2026), select $t_{3k-2} = 2.16k - 1.75$, $t_{3k-1} = 2.16k - 0.84$, and $t_{3k} = 2.16k$. For the synchronous impulse-switching sequence inspired by Gao et al. (2020), set $t_{2k-1} = 1.28k - 0.71$ and $t_{2k} = 1.28k$. The number of impulses N_I , the average impulsive interval T_I , and the performance cost J under different mechanisms are presented in Table 1. It can be observed that the proposed MDETM (20) requires fewer impulse executions than the compared methods while maintaining a comparable level of accumulated synchronization error.

Based on the synchronization result established in Theorem 3, an image encryption and decryption application is further presented, in which the synchronized states of the drive-response systems are used to generate consistent chaotic key sequences for encryption and decryption. The encryption procedure for a color image of size $p \times q \times 3$ is described as follows. (i) Preprocess the Image. Read the pixel matrices $\mathcal{M}_l(r, c)$, where $r = 1, 2, \dots, p$, $c = 1, 2, \dots, q$, and $l = 1, 2, 3$. Convert the matrices

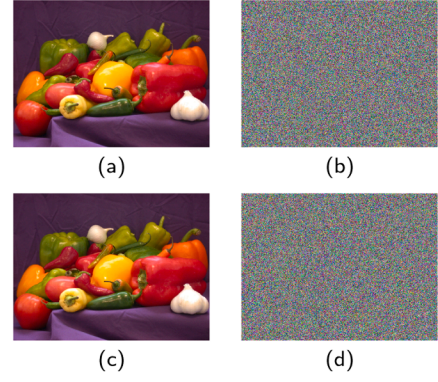


Fig. 10. (a) Original image. (b) Encrypted image. (c) Decrypted image with correct key. (d) Decrypted image with wrong key.

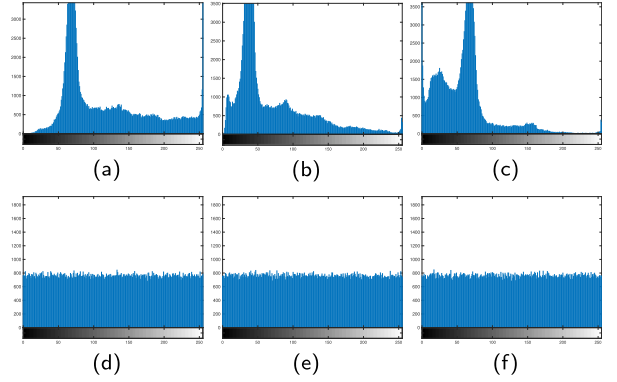


Fig. 11. Histograms of RGB components. (a)-(c) Original image. (d)-(f) Encrypted image.

$\mathcal{M}_l(r, c)$ into $1 \times pq$ sequences $\mathcal{O}_l(s)$, where $s = 1, 2, \dots, pq$. (ii) Generate the Chaotic Sequences. Once synchronization between systems (14) and (15) is achieved, three chaotic sequences are generated as

$$C_1(s) = 10^{11} \times (x_1(s) - \lfloor x_1(s) \rfloor) \bmod 256,$$

$$C_2(s) = 10^{11} \times (x_2(s) - \lfloor x_2(s) \rfloor) \bmod 256,$$

$$C_3(s) = 10^{11} \times (x_3(s) - \lfloor x_3(s) \rfloor) \bmod 256,$$

where $x_3(s) = \frac{x_1(s) + x_2(s)}{2}$, $\lfloor \cdot \rfloor$ denotes the floor operation, and $\bmod(\cdot)$ is the modulo operation. (iii) Diffusion Process. The diffusion operation is performed according to Algorithm 2, where $\oplus$ denotes the bitwise XOR operation. The pixel matrices of the encrypted image are then obtained by converting the sequences $\mathcal{N}_l(s)$ into $p \times q$ matrices. The decryption procedure, being the inverse process of encryption, is omitted.

Next, a series of metrics is employed to assess the performance of our encryption algorithm (Li et al., 2021). (i) Key Space and Key Sensitivity. The key space consists of the system matrices for two modes, the initial functions, the triggering parameters, and the decay rate. Theoretically, its size is sufficient to withstand brute-force attacks. Fig. 10 presents the encrypted and decrypted results for the color image Pepper. As illustrated, even a minute alteration to the encryption key ($x_{i0} = \psi_1 + (10^{-10}, 0)^T$) results in an entirely different decrypted image, indicating the high key sensitivity of the algorithm. (ii) Image Histogram. As depicted in Fig. 11, the histograms of the encrypted image approximate a uniform distribution. (iii) Correlation Analysis. Table 2 lists the correlation coefficients of adjacent pixels along horizontal, vertical, and diagonal directions, demonstrating a weak correlation in the encrypted image. (iv) Information Entropy. Table 3 shows that the entropy values of the encrypted image closely approach the theoretical ideal of 8, reflecting its strong randomness. (v) Resistance to Differential Attacks. As presented in Table 4, the number of pixel change rate (NPCR)

Table 2
Correlation coefficients of adjacent pixels.

Component	Direction	Original image	Encrypted image
R	H	0.9943	-0.0039
	V	0.9903	-0.0015
	D	0.9854	-0.0013
G	H	0.9908	0.0002
	V	0.9852	-0.0004
	D	0.9778	0.0019
B	H	0.9871	-0.0020
	V	0.9783	-0.0015
	D	0.9695	-0.0001

Table 3
Information entropy.

Component	Original image	Encrypted image
R	7.1853	7.9991
G	7.0237	7.9991
B	6.7915	7.9990

Table 4
NPCR and UACI of Pepper.

Component	NPCR (%)	UACI (%)
R	99.5033	33.4011
G	99.5014	33.4223
B	99.3362	33.3492

and unified average changing intensity (UACI) approximate their theoretical ideals (99.6094% for NPCR and 33.4635% for UACI). The above analysis confirms the validity and robustness of the encryption algorithm.

Remark 7. Unlike existing synchronization-based encryption algorithms (Long et al., 2023; Qi et al., 2024), this work does not require the permutation procedure while still effectively disrupting the correlation of adjacent pixels. As a result, it significantly reduces computational complexity and memory consumption.

6. Conclusion

This paper established several Razumikhin-type criteria to guarantee GES of nonlinear switched delay systems under ETIC. A novel MDETМ was proposed that effectively overcomes the adverse effects of time delay under the switching topology, while also ruling out Zeno behavior. The theoretical results were further extended to the exponential synchronization of SNNs, in which the desired MDETМ and impulsive control gain can be directly obtained using LMIs. The validity of the developed ETIC strategy was demonstrated through two numerical examples, including an application to image encryption. In view of the inevitable sampling and transmission delay of impulse signals, future work will aim to develop event-triggered delayed impulsive control strategies for switched systems. Moreover, additional factors such as external disturbances will also be considered in subsequent research.

CRedit authorship contribution statement

Zhifeng Lu: Writing – original draft, Software, Formal analysis;
Haiying Wang: Writing – review & editing, Validation;
Yujuan Tian: Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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